Midterm 1

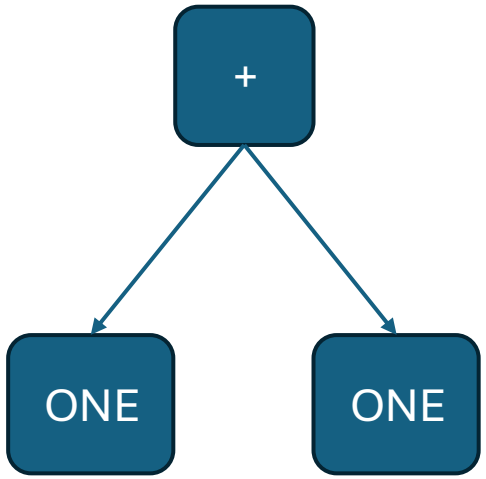
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Architecture

- Input
- ↓ Source AST
- ↓ Visitor Compiler
- ↓ Church-encoded lambda
- ↓ Call-by-value evaluator
- ↓ Observer
- Value

Abstract Syntax Trees

- D0Eop2(op, e1, e2)
- D0Eop2((+), ONE, ONE)



Church encoded-lambda

- Term ::= Var(name) | Lam(name, Term) | App(Term, Term)
- Term ::= x | $\lambda x.\, e$ |$e_1 e_2$

How do we represent primitives?

- TRUE = $\lambda x.\ \lambda y.\ x$
- FALSE = $\lambda x.\ \lambda y.\ y$

- 0 = $\lambda f.\ \lambda z.\ z$ (ZERO)
- 1 = $\lambda f.\ \lambda z.\ f\ z$ (ONE)
- 2 = $\lambda f.\ \lambda z.\ f\ (f\ z)$ (TWO)
- 3 = $\lambda f.\ \lambda z.\ f\ (f\ (f\ z))$ (THREE)

- Pair A B = $\lambda a.\ \lambda b.\ \lambda p.\ p\ a\ b$

Call by value

- let f x = x + 1; f (5+5)
- $\lambda x.\, x\ ((\lambda y.\, y)\ k)$

Observer

- 3 = $\lambda f.\lambda z.f\ (f\ (f\ z))$ (THREE)
- z is the base

Visitors

- If (op = “+”) then … else …
- Accept (visitor, self)

- AddVisitor:
  - IntAddVisitor()
  - BoolAddVisitor()